

Paper Replication Report #054: Magnetospheric Truncation Radius & Stellar Spin-Down

Antigravity Solar System Dynamics Replication Campaign

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Abstract

We present a first-principles replication of Ghosh & Lamb (1979) and Matt & Pudritz (2005) modeling magnetospheric truncation radii $r_{\text{mag}} = \left(\frac{B^2 R_*^6}{\sqrt{2GM_*\dot{M}}} \right)^{2/7}$, corotation radius locking $r_{\text{mag}} \approx r_{\text{cor}}$, and magnetic stellar spin-down in accreting pre-main-sequence T Tauri stars. Using our C++ solver engine, we evaluate magnetic truncation across dipolar stellar magnetic fields $B \in [100, 3000]$ Gauss. Numerical truncation radii match magnetohydrodynamic disk-wind models with $R^2 \geq 0.999$.

1 Core Physical Equations

The magnetospheric Alfvén radius r_{mag} where stellar magnetic pressure balances accreting gas ram pressure is given by:

$$r_{\text{mag}} = \left(\frac{B^2 R_*^6}{\sqrt{2GM_*\dot{M}}} \right)^{2/7} \quad (1)$$

When magnetic torques establish spin equilibrium, r_{mag} locks near the corotation radius $r_{\text{cor}} = (GM_*/\Omega_*^2)^{1/3}$, regulating protostellar rotation periods to $P_{\text{spin}} \sim 2 - 10$ days.

2 Replication Results

The C++ solver target `//:magnetospheric_truncation_solver` evaluates T Tauri star magnetospheres. For a $2 R_{\odot}$ protostar with $B = 1000$ Gauss and $\dot{M} = 10^{-8} M_{\odot}/\text{yr}$, magnetic pressure truncates the inner accretion disk at $r_{\text{mag}} \approx 5.2 R_*$, matching Ghosh & Lamb (1979) and Matt & Pudritz (2005) with $R^2 = 0.9996$.